

c) I = indicator that problem is correct.

X_k = indicator that k is correct, $k=1,2,3$

$X_k \sim \text{Bernoulli}(0.6)$

$$\begin{aligned} P(S=1) &= P(\text{success on attempt } 1,2,3) \\ &= P(X_1=1) + P(X_1=0, P(X_2=1)) + P(X_1=0, \\ &\quad X_2=0, X_3=1) \end{aligned}$$

Because they are independent, we just plug in:

$$= 0.6 + (0.4)(0.6) + (0.4)(0.4)(0.6)$$

$$= 0.6 + 0.24 + 0.096 = 0.936$$

$$E_1 = \frac{S}{20} \quad S \sim \text{Bernoulli}(0.936)$$

$$E[E_1] = E\left[\frac{S}{20}\right] = \frac{1}{20} E[S] = \frac{0.936}{20} = 0.0468$$

Variance:

$$\text{Var}(E_1) = \text{Var}\left(\frac{S}{20}\right)$$

$$= \frac{1}{400} \text{Var}(S)$$

$$= \frac{1}{400} (0.936)(0.064)$$

$$= \frac{0.059904}{400}$$

$$= 0.00014976$$